Discussion Questions

1

Two identical *metal* balls are suspended by equal length insulating strings. Both balls have the same net charge. In this problem, do not assume the balls are point charges.

(a) Draw separate free-body diagrams for each ball. Label the forces to indicate:

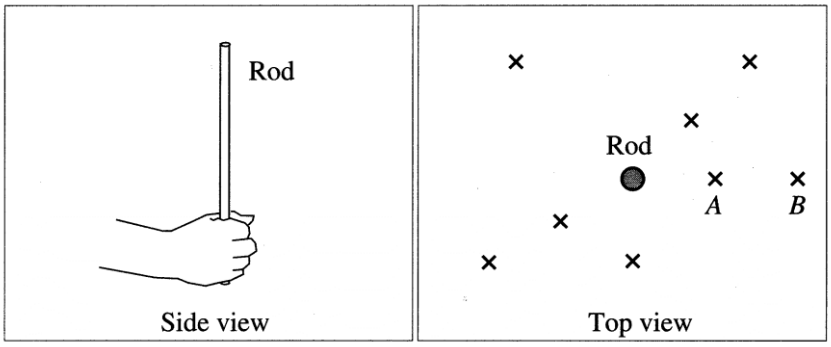
- The object exerting the force,
- The object on which the force is exerted
- The type of force (gravitational, normal, friction, etc.)

(b) Predict what will happen if the charge on the second ball is reduced slightly, so it is less than that on the first ball. Draw a sketch to illustrate your answer.
Is the angle that ball 1 makes with the vertical, *greater than, less than,* or *equal to* the angle that ball 2 makes with the vertical? Explain your reasoning.

(c) Predict what will happen if the net charge on the second ball is reduced to zero. Draw a sketch to illustrate your answer.

2

A charged acrylic rod and a small ball (a test charge) carries a charge q_{test} of the *same sign* as the charge on the rod. (The rod should be considered to be 'long' – remember, electric fields from an ~infinite rod are different to from a point charge) The top view figure shows 8 locations where the ball will be placed, marked with an 'x'.



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(a) Sketch *vectors* at each of the eight marked points to represent the electric force on the ball at that point.

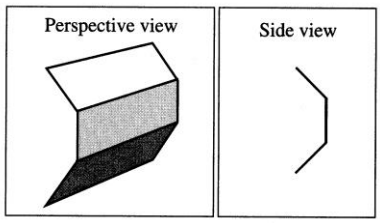
(b) How does the magnitude of the force exerted on the ball at point *A* compare to the magnitude of the force on the ball at point *B*? (Take distances from the diagram.)

(c) Suppose that the charge, q_{test} on the ball were halved, while the charge on the rod remained unchanged. Would the electric force on the ball at each point change? If so, how? If not, explain why not.

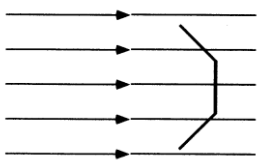
(d) Would the magnitude or the direction of the electric field at point *A* change if:

- The charge on the *rod* were increased? Explain.
- The magnitude of the charge on the ball were increased? Explain.
- The sign of the charge on the ball were changed? Explain.

3



(a) A piece of paper is folded into three portions of equal area as shown in the diagram.
How many area vectors are needed to describe the surface of the paper? Sketch the area vectors on the side view diagram and label them $\mathbf{A_1}$, $\mathbf{A_2}$, etc.



(b) Consider an imaginary surface in a uniform electric field $\boldsymbol{E}$ as shown. The surface has the same size and shape as the paper above. Is the flux through the top third of the surface *greater than, less than,* or *equal to* the flux through the middle third? Explain.

(c) Write an expression for the net electric flux Φ_{net} through the surface in terms of the area vectors and the electric field $\boldsymbol{E}$.

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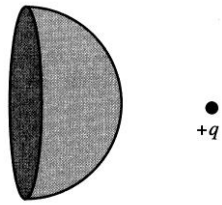
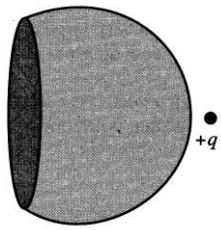
4



A *closed* Gaussian surface consists of a hemispherical surface and a flat plane as shown. A point charge *+q* is placed outside the surface, and no charge is enclosed by the surface.

(a) State the flux through the *entire* closed surface. Explain.

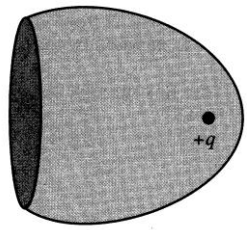
(b) Let Φ_L represent the flux through the flat left-hand portion of the surface. Write an expression in terms of Φ_L for the flux through the curved portion of the surface, Φ_C.



(c) Suppose that the curved portion of the Gaussian surface in part (a) is replaced by the larger curved surface as shown. The flat left-hand portion of the surface is unchanged.

(i) Does the value of Φ_L change? Explain.

(ii) How does the flux through the new curved portion of the surface compare to the flux through the original curved portion of the surface? Explain.

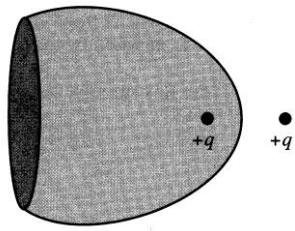


(d) Suppose that the curved portion of the Gaussian surface is replaced by a larger curved surface that encloses the charge as shown in the picture at right. The flat left-hand portion of the surface is still unchanged.

(i) Does the value of Φ_L change? Explain.

(ii) How does the flux through the new curved portion of the surface compare to the flux through the original curved portion of the surface? Explain.

(iii) Use Gauss' law to write an expression in terms of Φ_L and q for the flux through the curved portion of the surface.



(e) A second point charge *+q* is placed to the right of the Gaussian surface as shown.

(i) Is the value of Φ_L *greater than, less than,* or *equal to* the value of Φ_L in part (c)? Explain.

(ii) Is the value of the flux through the entire Gaussian surface *greater than, less than,* or *equal to* the value of the flux through the entire Gaussian surface in part (d)? Explain.

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5

A solid, spherical conductor is given a non-zero net charge. Which one of the following statements is true: The electrostatic potential of the conductor is

a) largest at the centre
b) largest on surface
c) largest somewhere between centre and surface
d) constant throughout volume

Problem-solving Questions

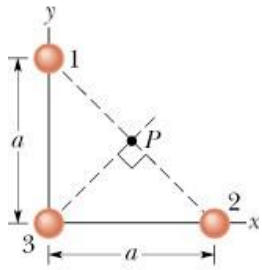
6

A point charge ($+3.00 \times 10^{-6}$C) is 12.0 cm from a second point charge (-1.50×10^{-6} C). Find the magnitude and direction of the force on each charge. Show the direction on a diagram.

7

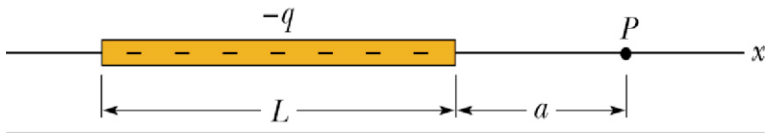
In the diagram, three particles are fixed in place.
They have charges $q_1 = +q$, $q_2 = +q$ and $q_3 = +2q$. $a = 6.00$ µm.

Find an expression for the magnitude and direction of the net electric field at *P*.



8 (mini-challenge question)

The diagram shows a *non-conducting* rod of length L has charge $-q$ uniformly distributed along its length.



(a) State the linear charge density of the rod (include units).
(b) Calculate the electric field at point P, a distance a from the end of the rod. *NOTE* an integration is required. (Hint: use Coulomb's Law for every charge element dq length dL)
(c) If P were *very far* from the rod compared to L ($a >> L$), the rod would look like a point charge. Show that your answer to (b) reduces to the electric field of a point charge in this case

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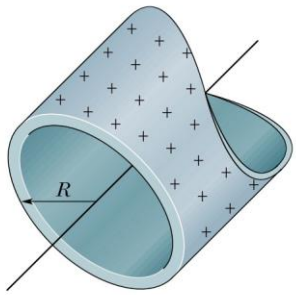
9

A point charge of 1.8 μC is at the centre of a *cubical* Gaussian surface with a 55 cm edge. What is the net electric flux through *one* side of the cubical surface?

10

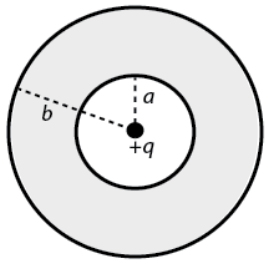
The figure shows a section of a long, thin-walled metal (conducting) tube of radius R with a charge per unit length = λ C/m on its surface. Derive expressions for E in terms of the distance r from the tube axis, considering both



(a) $r > R$, and
(b) $r < R$. Plot your results for the range $r = 0$ to $r = 5.0$ cm, assuming that $\lambda = 2.0 \times 10^{-8}$ C/m and $R = 3.0$ cm.

11

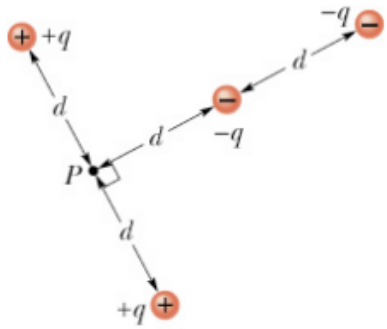
A point charge $+q$ is placed at the centre of an electrically neutral, spherical conducting shell with inner radius a and outer radius b.
What charge appears on (a) the inner and (b) the outer surfaces of the shell?
What is the net electric field at a distance r from the centre of the shell if
(c) $r < a$
(d) $b > r > a$, and
(e) $r > b$?
Sketch field lines for those three regions.
For $r > b$ what is the net electric field due to
(f) the central point charge,
(g) the inner surface charge and
(h) the outer surface charge?



Q 9, 10 & 11 *From Fundamentals of Physics, 7/e, Resnick, Halliday and Walker, © John Wiley 2005. Reproduced with permission.*

12

From the figure below, calculate the net potential at point P due to the four point charges. Take $V = 0$ at infinity, $q = 5.00$ fC and $d = 4.00$ cm? (fC = 10^{-15} C)



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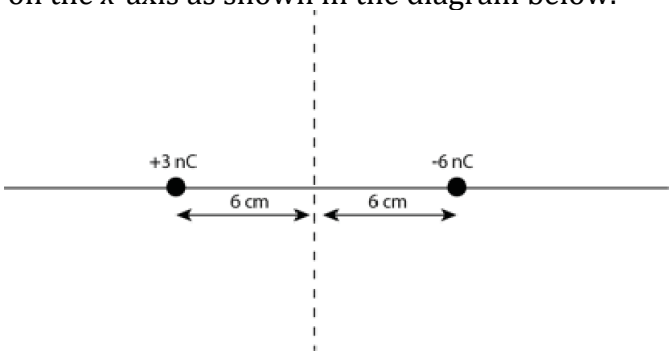
13 (mini-challenge)

Two charges are placed on the *x*-axis as shown in the diagram below.



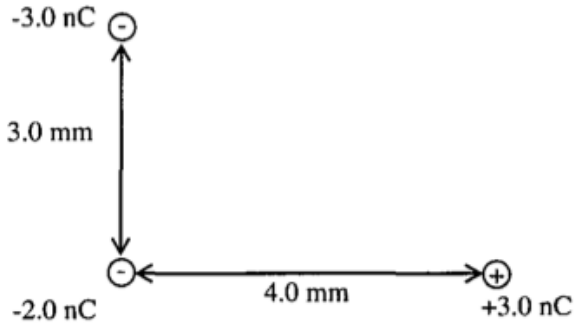
(a) Find any points of zero potential anywhere on the *x*-axis.
(b) Do these points (if any) correspond with points of zero electric field on the *x*-axis?
(c) Are there any points off the *x*- axis where the potential is zero?
(d) Are there any equipotential lines (lines of equal potential) in the *x*-*y* plane? (qualitative answer only)

Past exam questions

14

(a) Two point particles carry charges of +3.0 nC and -2.0 nC and are separated by a distance of 4.0 mm.

(i) What is the magnitude and direction of the force on the positive charge caused by the negative charge?
(ii) What is the magnitude and direction of the force on the negative charge caused by the positive charge?
(iii) A third charge of -3.0 nC is placed a distance of 3.0 mm from the negative charge as shown in the figure. The three charges form the corners of a right-angled triangle. What is the net force on the charge +3.0 nC due to the other two charges?



(b) When a plastic comb is rubbed with wool, the comb becomes negatively charged. It is observed that small particles such as dust carrying no net charge are attracted to the comb. In one to two pages, using diagrams where appropriate, explain why this happens. In your answer you should address the following points:

- Why does the plastic comb become charged?
- How is the charge distribution within the electrically neutral particles affected by the electric field of the comb?
- Why are electrically neutral particles attracted to the comb?
- Why is it important that the field produced by the comb is not uniform for the neutral particles to be attracted to the comb?

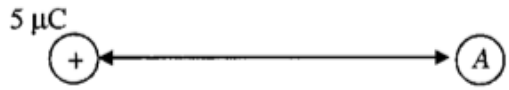
15

(a) A glass rod is charged to +5 nC by rubbing.

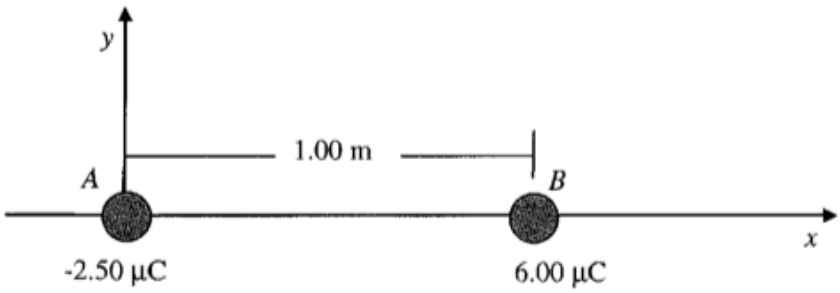
(i) Have electrons been removed from the rod or protons added?

(ii) How many particles have been removed or added?

(b) A small charged particle (A) experiences an electrostatic force of magnitude 5×10^{-2}N toward a small particle carrying a charge of +5.0 μC which is a distance of 10.0 cm from it. Given that there are no other charged particles in the vicinity of these two, what is the charge carried by the particle A?



(c) Consider two charges lying on the x-axis as shown in the figure. A charge of -2.50 μC (charge A) is located at x = 0.00 m and a charge of +6.00 μC (charge B) is at x = +1.00 m.



(i) What is the direction of the electric field in the region between the two particles, A and B?

(ii) What is the direction of the electric field on the x-axis to the right of particle B?

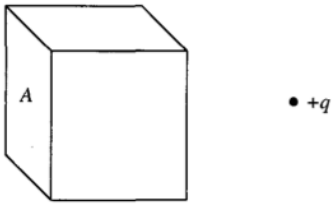
(iii) What is the direction of the electric field on the x-axis to the left of particle A?

(iv) Determine the point on the x-axis (other than at infinity) at which the electric field is zero.

16

(a) Consider the *closed* surface of the cube where one of the surfaces is labelled *A.* An electric charge $+q$ is located outside the cube as shown in the figure.

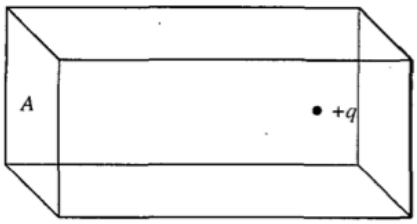
(i) What is the total electric flux through the *entire* surface of the cube? Explain.



Suppose the cube is extended so that it is now a rectangular prism and it completely encloses the charge q. The position of the surface A relative to the charge is unchanged.

(ii) Does the value of the flux through surface A, denoted Φ_A, change? Explain.

(iii) Use Gauss' law to write an expression in terms of Φ_A and q for the flux through the

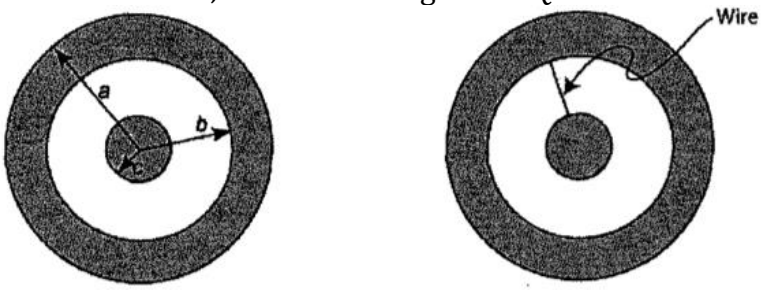


remainder of the surface.

(b) A large flat *conducting* sheet carries a surface charge density of $+\sigma$ C/m^2 across its total surface. Using Gauss' law, determine an expression for the magnitude and direction of the electric field near the surface of the conducting sheet. Carefully explain all the stages of your calculation.

17

Consider two concentric conducting spheres (as shown in the figure). The outer sphere is hollow, with an inner radius of b and an outer radius of a, and initially given a charge of $-7Q$. The inner sphere is solid, and has a radius of c, and has a charge of $+2Q$ on it.



(i) Explain why the electromagnetic field inside the conducting regions, $b < r < a$ and $r < c$, must be zero, where r is the distance from the centre of the inner sphere.

(ii) Using Gauss' law, find the total charge lying on the inner surface of the hollow conductor (i.e. at $r = b$).

(iii) How much charge lies on the outer surface of the hollow conductor (i.e. at $r = a$)?

(iv) Using Gauss' law, find an expression for the electric field outside the spherical shell, i.e. for $r > a$. Explain all steps in your derivation

[Hint: the surface area of a sphere of radius R is $4\pi R^2$.]

Suppose a conducting wire is then connected between the inner and outer spheres, as in the figure on the right.

(v) After electrostatic equilibrium is reached, how much total charge is on the outside sphere?

(vi) How much charge now lies on the inner surface of the hollow conductor?

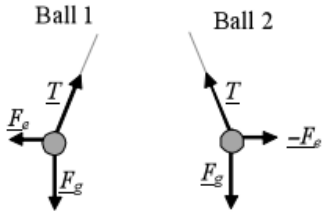
(vii) How much charge now lies on the outer surface of the hollow conductor?

Physics 2: Physical Science & Technology (PHYC10004) SUM 2025

Short solutions

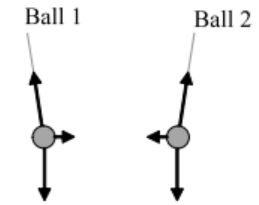
Discussion Questions:

1a



1b Angle reduces but balls make same angle with vertical as each other

1c



They attract because the charged ball induces separation of charges on the uncharged ball. *(If they were to touch they would share the available charge, so each would get $q/2$- and then they would repel)*

2a forces are radially outwards from the position of the rod

2b force on A about twice the force on ball at B (E field is not inversely proportional to R^2; closer to inversely proportional to R)

2c would be halved **2d** field would increase ($\propto Q$ on rod); field would not change; No, does not affect the field as it is the charge on the *rod* that establishes the field..

3 flux through top and bottom thirds less than through the middle third;

$F_{net} = E \cdot A_1 + E \cdot A_2 + E \cdot A_3$ (without detailed cos values, clearly $E.A_2$ is larger than the others).

4a total flux = 0, due to the charge outside the closed surface, all field lines must pass in and out again

4b F_L must be equal to $-\Phi_C$ for the total flux to be zero

4ci F_L does not change; it's the same distance away and the same shape, the integral giving the flux must be the same.

4cii flux does not change; total flux is still zero

4di F_L does not change; it's the same distance away and the same shape, the integral giving the flux must be the same.

4dii & iii This will now change; and instead of $F_L = -\Phi_C$ we will have $F_C = \frac{q}{e_0} - F_L$

4ei greater than; integral will now be larger

4eii it will be the same as before; the enclosed charge is the same

5 answer is (d) - no field inside a conductor, hence no work required to move charges around)

Problem Solving Questions:

6 2.81 N, attractive. **7** $E_{net} = q/\pi\varepsilon_0 a^2$ **8a** charge density (λ) = $-q/L$ C/m

8b $E_x = -\frac{1}{4pe_0}\left(\frac{q}{a(L+a)}\right)$

9 3.4×10^4 N m^2 C^{-1} **10a** $E = \lambda/(2\pi\varepsilon_0 r)$ **10b** $E = 0$ **Note:** $E_{max} = 1.2 \times 10^4$ N C^{-1}.

11a&b $-q$ on the inner surface, $+q$ on the outer surface **c** $E=q/(4\pi\varepsilon_0 r^2)$ radially outwards **d** still zero inside the conductor **e** $E=q/(4\pi\varepsilon_0 r^2)$ radially outwards all field lines are straight lines along radii (except where $E = 0$) **f &g** $E=0$ **h** $E=q/(4\pi\varepsilon_0 r^2)$ radially outwards

12 5.6×10^{-4} V

13a two points: 4 cm to the right of the 3 nC charge and 12 cm to the left of the 3 nC charge
b No **c** Yes; the condition is that the distance to the 3 nC charge must be half the distance to the 6 nC charge. For example, there is a point at (-6, +4√3).
d lots of them! (eg the 0 V equipotential line must go through (-4, 0) and (-18, 0); shape can be inferred from field lines (perpendicular to these)

Past Exam Questions:

14 (2005)

(a)i. 3.4×10^{-3} N; attractive (to the left); **ii.** same magnitude, to the right; **iii.** vector combination gives $-5.97\mathbf{i} + 1.94\mathbf{j}$ mN.
(b) Transfer of electrons from the comb to the wool; some polarization of these particles; polarization of the neutral particles leads to unequal repulsive and attractive forces, with the attractive force the stronger (needs a diagram); if the force was uniform all it would do would be to rotate the (polarized) particles, whilst in a non-uniform field there is also a net attraction,...

15 (2006)

ai electrons removed **aii** 3.1×10^{10} electrons **b** -11 nC
ci to the left **cii** to the right **ciii** to the right and then the left
civ 1.82 m to left of A

16

(**a**) **i.** Zero (Gauss' law); **ii.** no – same geometry as before; **iii.** $\Phi_{\text{Remainder}} = q/\varepsilon_0 - \Phi_A$
(**b**) In the text (HRW 11th Ed, §23.3-5, pp669-675) (Gaussian *through* surface, $E = \sigma/2\varepsilon_0$)

17

(**i**) no E inside conductors (**ii**) $q = -2Q$ (**iii**) $q' = -5Q'$
(**iv**) $E(r) = (1/4\pi\varepsilon_0)(-5Q/r^2)$ (**v**) all = $-5Q$ (**vi**) 0 (vii) $-5Q$